

■ ChipPilot AI — Engineering Audit Report

[VIOLATIONS DETECTED]

Target Module: cpu_top | Generated: 2026-08-23 11:09:41

Engine: v1.0.0 (EDA Correlated)

1. Executive PPA & Diagnostic Summary

Worst Slack (WNS)	Synthesized Cells	Inferred Latches	Static Lint Rules	Ranked RCA Causes
-2.04 ns Target 10.00 ns	0 cells Yosys Logic Synth	0 Sequential Hazard	0 issues Verilator Static	6 hypotheses Multi-Factor Score

2. Static Timing Analysis (STA) Critical Paths

Startpoint	Endpoint	Slack (ns)	Arrival (ns)	Required (ns)	Status
alu.a[31]	alu.result[31]	-2.040	11.840	9.800	VIOLATION

3. Ranked Root Cause Analysis & Diagnostic Hypotheses

#1 Critical Setup Timing Violation (alu.a[31] -> alu.result[31]) Location: alu:1 Diagnostic Hypothesis: Excessive combinational logic depth between registers exceeds the clock period constraint. Multi-Tool Corroborated Evidence: [OpenSTA:Timing] Path: alu.a[31] -> alu.result[31] [OpenSTA:Timing] Data Arrival: 11.8400 ns, Data Required: 9.8000 ns [OpenSTA:Timing] Setup Slack: -2.0400 ns (VIOLATION)	Confidence: 92% Severity: ERROR
#2 Structural Hazard: INCOMPLETE_CASE_STATEMENT in alu Location: alu.v:18 Diagnostic Hypothesis: Combinational case statement lacks 'default' clause. High risk of unwanted latch inference. Multi-Tool Corroborated Evidence: [AST Scanner:alu.v:18] INCOMPLETE_CASE_STATEMENT: Combinational case statement lacks 'default' clause. High risk of unwanted latch inference.	Confidence: 51% Severity: WARNING
#3 Structural Hazard: MISSING_ASYNC_RESET in regfile Location: regfile.v:21 Diagnostic Hypothesis: Sequential always block does not include reset logic. Registers will power-up in undefined state. Multi-Tool Corroborated Evidence: [AST Scanner:regfile.v:21] MISSING_ASYNC_RESET: Sequential always block does not include reset logic. Registers will power-up in undefined state.	Confidence: 51% Severity: WARNING

4. Synthesis & RTL Static Quality

- **Gate Cell Statistics:** Total 0 synthesized standard cells (Standard Logic Gates).
- **Sequential Integrity:** 0 inferred asynchronous latches detected.
- **Static Lint Violations:** 0 warnings/errors reported by Verilator.

Flagged AST Structural Hazards:

Module	Severity	Hazard Type	Line	Description
alu	WARNING	INCOMPLETE_CASE_STATEMENT	18	Combinational case statement lacks 'default' clause. High risk of unwanted latch inference.
regfile	WARNING	MISSING_ASYNC_RESET	21	Sequential always block does not include reset logic. Registers will power-up in undefined state.
control_unit	WARNING	INCOMPLETE_CASE_STATEMENT	20	Combinational case statement lacks 'default' clause. High risk of unwanted latch inference.
control_unit	INFO	UNUSED_NET	0	Declared register 'unused_config_reg' is never referenced in module logic.
data_memory	WARNING	MISSING_ASYNC_RESET	15	Sequential always block does not include reset logic. Registers will power-up in undefined state.

5. Prescribed PPA Optimization Patches

Category	Action / Target	Prescription & Verification	Expected Gain
LATCH_ELIMINATION	ADD_DEFAULT_ASSIGNMENT demo_rtl/faulty/alu.v	Add default case assignment in ALU always block to eliminate latch inference. Verify: Re-run Verilator lint and Yosys synthesis	Eliminates transparent latch \$DLATCH_P_

6. Developer Verification & Engineering Sign-Off

This section is reserved for developer review, closed-loop regression verification, and formal sign-off.

Verification Checklist Item	Automated Status	Developer Sign-Off Note / Signature
1. Static Timing Constraints (Setup/Hold WNS >= 0.0ns)	FAIL	_____
2. Asynchronous Inferred Latch Elimination	CLEAN	_____
3. Verilator Static Rule Conformance	CLEAN	_____
4. Logic Equivalence & Closed-Loop Regression	PENDING QA	Reviewer: _____ Date: _____